QP CODE: 25900870
MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025
FIRST SEMESTER
Discipline Specific Core Course (DSC) - MG1DSCIAM100 - INTERACTIVE ROBOTIC SYSTEMS
(2024 ADMISION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Part A
Multiple Choice Questions
Answer All Questions
Each Question carries 1 mark

1. a). pin10 / b). pin11/ c). pin12 / d). pin13 [CO1]
2. a). generate a tone on a pin [CO1]
3. c). C/C++ [CO1]
4. a). Infrared Radiation [CO2]
5. b). resistance [CO2]
6. b). 4 [CO2]
7. a). Feedforward Control [CO3]
8. c). 8 [CO3]
9. b). 0-180 degrees [CO3]
10. d). All of the above [CO3]

Part B
Short answer Type Questions
Answer any 4 Questions
Each Question carries 5 marks

11. Check the void setup() function - Check for infinite loops in setup() - Ensure the code compiles without errors in the Arduino IDE- Ensure correct board/port selection - Look for blocking functions or delay. [CO1]

12. + Addition, -Subtraction, *Multiplication, / Division, %Modulus (remainder), ++ Increment, -- Decrement. [CO1]

13. Control the direction of a DC motor - The purpose of an H-bridge circuit is to allow a DC motor to rotate in both directions by controlling the polarity of the voltage applied to it. [CO2]

14. LDR changes resistance continuously. The resistance of an LDR decreases when light increases, and increases when light decreases. This produces a range of voltage values—not just 0 or 1. [CO2]

15. The HC-SR04 sensor is not suitable for outdoor use because rain, fog, dust, and moisture interfere with ultrasonic waves and can damage the non-waterproof electronics. Temperature and wind changes also affect the speed and direction of sound, leading to inaccurate or unstable distance readings in outdoor environments.[CO3]

16. Infrared (IR) sensors are devices that use infrared light to detect objects or measure distance. They work by emitting IR light from an LED and measuring the amount of reflected light received by a photodiode or phototransistor. When an object is nearby, more IR light is reflected back, allowing the sensor to detect its presence. This makes IR sensors commonly used for simple and fast obstacle detection. [CO3]

Part C
Essay Type Questions
Answer any 2 Questions, Each Question carries 10 marks

17. [CO1]

```
int led = 13;
int count = 0;
void setup() {
 pinMode(led, OUTPUT);
}
void loop() {
 count = 0;
 while (count < 5) {
  digitalWrite(led, HIGH);
 delay(500);
 digitalWrite(led, LOW);
delay(500);
count++;
 }
  while (1);
}
```

18. Stator (Field Magnet), Rotor / Armature, Commutator, Brushes, Armature Windings, Shaft [CO2]

19. [CO2]

```
#include <Servo.h>
Servo myservo;
int count = 0;
void setup() {
  myservo.attach(9);
 }
void loop()
{
  count = 0;
while (count < 5)            {
 for (int pos = 0; pos <= 180; pos++)
  {
 myservo.write(pos);
 delay(10);
 }
    for (int pos = 180; pos >= 0; pos--) {
    myservo.write(pos);
    delay(10);
  }
 count++;
}
while (1);
}
```

20. Terrain and environmental conditions greatly impact robotic navigation. [CO3]

- Slopes require more torque and stability control.
- Surface texture affects traction and steering accuracy.
- Obstacles require detection and real-time avoidance strategies.